MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FIRST SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

MG1DSCTXL100 – TEXTILE SCIENCE

Duration: 1.5Hrs Maximum Marks: 50

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer **All** Questions

Each question carries **1** mark

1. Lengthy fibres are [U] [1]
   a) Staple fibres b) Coarse fibres c) Fine fibres d) Filament fibres
2. Wool fibre is made up of [U] [2]
   a) Cellulose b) Vitamin c) Keratin d) Hydrocarbons
3. A number that indicates the fineness or diameter of a yarn. [U] [3]
   a) Count b) Denier c) Twist d) Density
4. Which of the following is a fancy weave? [U] [4]
   a) Plain b) Twill c) Pile d) Satin
5. Example for a regenerated fibre. [U] [1]
   a) Polyester b) Rayon c) Nylon d) Acrylic
6. Acetic acid is used in the manufacture of [U] [2]
   a) Nylon b) Cotton c) Rayon d) Polyester
7. Which pinning produces bulky two-layered yarns? [U] [3]
   a) Ring b) Rotor c) Hand d) Open-end
8. A type of matted fabric that consists of textile fibers condensed and pressed together. [U] [4]
   a) Braids b) Net c) Felt d) Needle-punched
9. Yarns that contain larger and usually looser fiber clusters than slub yarns [U] [3]
   a) Chenille b) Novelty c) Flake d) Ratiné
10. Fabrics made up of a mix of two or more different raw fibres. [U] [4]
    a) Doubled b) Plied c) Blended d) Mixed

[1 x 10 = 10]

Part **B**

Fill in the Blanks Questions

Answer **All** Questions

Each question carries **1** mark

11. Silk is an example of an _____ fiber. [U] [1]
12. The process of separating cotton fibers from seeds is known as _____. [U] [2]
13. In hand spinning, a _____is often used to help twist and align the fibers manually. [U] [3]
14. Drill is an example of _____fabric. [U] [4]
15. A ____ refers to fabrics where yarns of different types are used in the fabric construction. [U] [4]

16. _____ fibers are man-made fibres. [U] [1]
17. Flax is commonly used to make _____ fabric, which is cool and breathable. [U] [2]
18. _____ yarn has a looped or curly texture, often used to add bulk. [U] [3]
19. _____ weave involves weaving two layers of fabric simultaneously. [U] [4]
20. Nets are open, mesh-like fabrics created by _____ or knotting yarns. [U] [4]

[1 x 10 = 10]

Part **C**

Answer in One Word or One Sentence

Answer **All** Questions

Each question carries **1** mark

21. Give the key characteristic of cotton fibre. [U] [1]
22. Mention the property of silk that gives it a shiny appearance. [U] [2]
23. In hand spinning, what is often used to assist in twisting fibers? [U] [3]
24. Recall the structure of a plain weave fabric which makes it extremely strong. [U] [4]
25. Name the type of yarn made by twisting together two or more plied yarns. [U] [3]
26. What type of fiber is obtained from plants, animals, or minerals? [U] [1]
27. When burned, what odor does nylon emit? [U] [2]
28. What does a higher yarn count indicate? [U] [3]
29. List any variation of twill weave. [U] [4]
30. Which weave is known for its ability to create complex patterns in fabric? [U] [4]

[1 x 10 = 10]

Part **D**

Short Answer Questions

Questions **any five** Questions

Each question carries **2** marks

31. Differentiate between filament fibres and staple fibres. [U] [1]
32. How can silk and wool be identified using microscopic test? [U] [2]
33. Briefly explain yarn twist and yarn count. [U] [3]
34. Discuss extra warp and extra weft designs. [U] [4]
35. Write a short note on hand spinning. [U] [3]
36. Draw the design of 5 end satin and list its variations. [U] [4]

[5 x 2 = 10]

Part **E**

Long Answer Questions

Questions **any two** Questions

Each question carries **5** marks

37. Classify textile fibres according to its source of origin. [U] [1]
38. Discuss in detail about the processing of cotton fibre. [U] [2]
39. Explain the Ring spinning process with a neat flow chart. [U] [3]
40. Elaborate on any three non-woven fabric construction methods. [U] [4]

[2 x 5 = 10]